Journal of Science and Medicine in Sport 20 (2017) 781–785

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Original research

A sled push stimulus potentiates subsequent 20-m sprint performance

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ARTICLE INFO

Article history:
Received 24 March 2016
Received in revised form 2 December 2016
Accepted 15 December 2016
Available online 23 January 2017

Keywords:
Post-activation potentiation
Resisted sprinting
Running
Football
Performance
Physical conditioning

ABSTRACT

Objectives: The objective of this study was to examine the potentiating effects of performing a single sprint-style sled push on subsequent unresisted 20 m sprint performance.
Design: Randomized crossover design.
Methods: Following a familiarization session, twenty rugby league players performed maximal unresisted 20 m sprints before and 15 s, 4, 8 and 12 min after a single sled push stimulus loaded with either 75 or 125% body mass. The two sled push conditions were performed in a randomized order over a one-week period. The fastest sprint time recorded before each sled push was compared to that recorded at each time point after to determine the post-activation potentiation (PAP) effect.
Results: After the 75% body mass sled push, sprint time was 0.26 ± 1.03% slower at the 15 s time point (effect size [ES] = 0.07) but faster at the 4 (−0.95 ± 2.00%; ES = −0.22), 8 (−1.80 ± 1.43%; ES = −0.42) and 12 (−1.54 ± 1.54%; ES = −0.36) min time points. Sprint time was slower at all the time points after the 125% body mass sled (1.36 ± 2.36%–2.59 ± 2.90%; ESs = 0.34–0.64).
Conclusions: Twenty-meter sprint performance is potentiated 4–12 min following a sled push loaded with 75% body mass while it is impaired after a 125% body mass sled. These results are of great importance for coaches seeking to potentiate sprint performance with the sled push exercise.
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1. Introduction

Post-activation potentiation (i.e., PAP) is the acute enhancement of performance following a preload stimulus or conditioning activity (i.e., CA). Various sporting tasks can be potentiated with a recent meta-analysis indicating that the PAP effect is small for jump (effect size [ES] = 0.29), throw (ES = 0.26) and upper-body ballistic (ES = 0.23) tasks, and moderate for running sprint activities (ES = 0.51).[1]

Running sprint performance has generally been potentiated with traditional resistance-exercises including the back squat[2–4] and the power clean,[3] and with plyometric exercises.[5] However, two studies reported unchanged sprint times following a back squat CA in rugby players.[6,7] These discrepancies might be explained by the fact that the occurrence of PAP is mediated by the net balance between potentiation and fatigue[8] generated after the completion of the CA. Performance may improve if potentiation dominates fatigue, remain unchanged if fatigue and potentiation are at similar levels, or decrease if fatigue dominates. This balance between potentiation and fatigue is influenced by several variables including the characteristics of both the CA (i.e., volume, load etc.) and the individual (i.e., strength level, training age etc.) as well as the rest period between the CA and the performance task.[1] Additionally, this variability in PAP susceptibility when a back squat is used to potentiate sprint performance might be explained by the lack of movement specificity between sprinting and the back squat exercise. With this in mind, sprint-specific CAs including resisted sprinting using sleds have recently been employed in an attempt to potentiate subsequent sprint performance.[9,10] Whelan et al.[9] reported a potentiation effect on 10 m unresisted sprints for up to 10 min after three 10 m sled pulls loaded with 25–30% body mass. Conversely, Winwood et al.[10] failed to potentiate 15 m sprint performance with a 150% body mass sled pull stimulus. Collectively, these findings suggest that the load on the sled impacts the magnitude of sprint PAP and this must be carefully considered when designing interventions to exploit the PAP response.

Another resisted sprinting device that is often implemented into training programs is the sled push exercise. Similarly to the sled pull, the sled push is believed to share movement similarities with sprinting and may therefore have the potential to potentiate subsequent sprint performance. However, to the authors' knowledge no attempt has been made to determine whether sprint performance

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http://dx.doi.org/10.1016/j.jsams.2016.12.074
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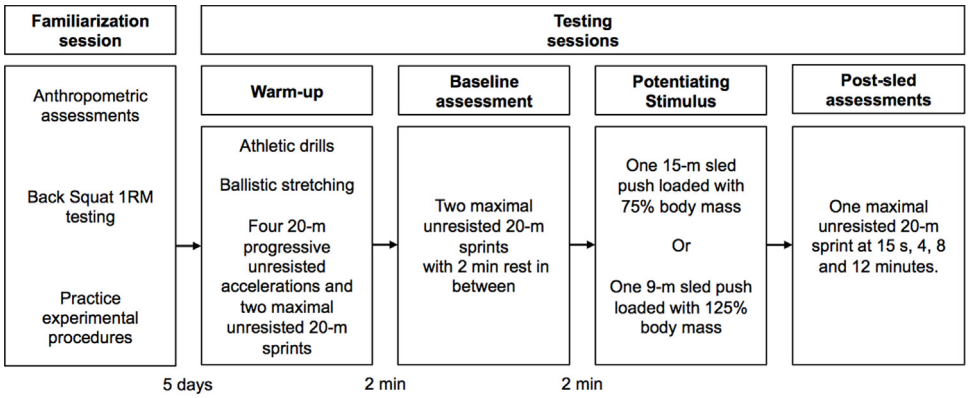

Fig. 1. Schematic representation of the study design. The experimental procedure consisted of two testing sessions each separated by seven days and performed in a randomized order.

is potentiated following a sled push conditioning activity. Moreover, no data is presently available regarding the influence of the load during a sled push on the magnitude of the PAP response to this activity. Therefore, the purpose of this study was to investigate the effects of performing a single sled push loaded with either 75 or 125% body mass on subsequent unresisted sprint performance.

2. Methods

Twenty rugby league players ([mean ± standard deviation] age: 18.4 ± 0.8 year; height: 180 ± 8 cm; body mass: 80.4 ± 6.8 kg) with an extensive heavy sled push and resistance training background (resistance training experience: 3.1 ± 1.8 year; number of weekly resistance training sessions: 3.0 ± 0.5) volunteered to complete one familiarization and two experimental sessions. They were informed of the aims, benefits, risks and procedures of the study before participating in the investigation. They were required to abstain from taking any stimulants or depressants prior to testing; this included abstention from caffeine for at least 6 h and alcohol for at least 24 h before testing. All players read and signed an informed consent form and those under 18 years of age had written informed consent from a guardian to participate in the study. All procedures were conducted in accordance with the Declaration of Helsinki and approved by the Edith Cowan University Human Research Ethics Committee (Project approval number: 10791). A schematic representation of the study design is presented in Fig. 1.

The familiarization session consisted of height and body mass measurements and familiarization with the experimental procedures. Height and body mass were measured with a calibrated Tanita stadiometer (model HR-200, Japan) and a Tanita scale (model BC-418MA, Japan), respectively. The intraclass correlation (ICC) value for height and body-mass measurements was 0.97 and 0.98, respectively. The players were then familiarized with the experimental procedure including 20 m sprints and sled pushes loaded with 75 and 125% body mass.

The two experimental testing sessions were separated by seven days and were completed in a randomized order and at the same time of the day, with the first session taking place three days after the familiarization session. On arrival, the players performed a standardized warm-up including three minutes of athletic drills (e.g. knee hug stretch, internal and external hip rotation, lateral shuffles, high knees, heel kicks, lunges and twist, and straight-leg march) and two ballistic stretching exercises (e.g. frontal and lateral leg swings, 10 repetitions on each leg). They were then required to complete four bursts of progressive unresisted accelerations over 20 m and one unresisted 20 m sprint with maximal effort with one minute of recovery between the sprints. Two minutes after the end of the warm-up, the participants completed two maximal unresisted 20 m sprints with two minutes of recovery in between (i.e., baseline sprints). After two minutes of rest, they completed a single sled push at either 75 or 125% body mass, followed by a maximal unresisted 20 m sprint that was performed 15 s, 4, 8 and 12 min after the sled push (i.e., post-sled). The following formula was used to equate sled push loads and distances:

$$Mechanical \quad work = load \times distance$$

The sled pushes at 75 and 125% body mass were therefore performed over 15 and 9 m, respectively. The following equation was used to determine the potentiating effect of performing a sled push on subsequent sprint performance:

$$\%PAP = \left[\left(Sprint_{(post-sled)} - Sprint_{(baseline)}\right) \div Sprint_{(baseline)}\right] \times 100$$

where, $Sprint_{(post\text{-}sled)}$ is the sprint time recorded at each time point (i.e., 15 s, 4, 8 and 12 min) after the sled push conditioning activity and $Sprint_{(baseline)}$ is the fastest sprint time recorded before the sled push.

The sprints were measured in an indoor environment using electronic timing gates (Microgate Photocell, Bolzano, Italy) positioned 20 m from the start line. All players initiated the sprint in their own time from a semi-crouched position with the front foot 20 cm from the start line. The timer started when the players broke the first pair of gates on the start line. The players received verbal encouragement to sprint at maximal effort. The intraclass coefficient correlation of the baseline 20 m sprint times was 0.95 (95% CI = 0.86–0.98).

The sled push was performed on an indoor artificial turf surface with the players required to accelerate the sled as quickly as possible over the prescribed distance. The hand position was not standardized and the players started in the position that was the most comfortable and effective for them. The only instruction was to keep the elbows flexed. All the players freely adopt a position where the superior portion of first web space between the thumb and index was positioned at approximately 50% of standing height on the vertical poles. This positioning was close to the one reported by Wu et al.[11] who found that 40% standing height body position produced the largest peak scrummaging forces in rugby players.

A two-way (Condition × Time) repeated measures ANOVA and effect sizes (ESs) were computed to quantify the change in sprint time at each time point (versus baseline sprint) after the two sled push conditioning activities. A post-hocg!> Bonferroni test was used when significant interaction effects were detected to determine which measures differed significantly. The statistical power for Condition and Time was 0.72 and 0.98, respectively.

Additionally, Cohen's d was also computed to determine the magnitude of difference in PAP at each time point between the two

Table 1
20-m sprint times before (±standard deviation) after the 75- and 125% body mass sled pushes and corresponding effect sizes (ES).

		Baseline	15 s	4 min	8 min	12 min
75% Sled	Time (s) ± SD	3.28 ± 0.14	3.29 ± 0.16	3.25 ± 0.18	3.22 ± 0.17*	3.23 ± 0.17*
	ES	–	0.07	−0.22	−0.42	−0.36
	ES interpretation		Trivial	Small	Small	Small
	Outcome[a]		Almost certainly not (0.3%)	Possibly (54.0%)	Very likely (97.8%)	Likely (94.0%)
125% Sled	Time (s) ± SD	3.30 ± 0.14	3.39 ± 0.20*	3.37 ± 0.19*	3.36 ± 0.20	3.35 ± 0.19
	ES	–	0.64	0.53	0.41	0.34
	ES interpretation	–	Medium	Medium	Small	Small
	Outcome[b]		Very likely (99.5%)	Very likely (98.6%)	Likely (90.9%)	Likely (90.6%)

Negative values represent positive changes to sprint time (i.e. faster sprint time) compared to baseline value. SD = standard deviation.
*Significantly different from baseline ($p \leq 0.05$).
[a] Likelihood that the effect is beneficial.
[b] Likelihood that the effect is harmful.

sled push conditioning activities. The magnitudes of both the ESs and Cohen's d were considered trivial (<0.20), small (0.20–0.50), medium (0.50–0.80) large (0.80–1.30) or very large (>1.30).[12] The chances that the true effect was beneficial, trivial or harmful were calculated on log-transformed data using a spreadsheet.[13] Briefly, this spreadsheet uses the p-value of the outcome measure and the smallest worthwhile change (calculated as one-fifth of the between subject standard deviation[14]) to determine the likelihood that the true magnitude of the effect is substantial in a clinically beneficial, trivial or harmful way. The smallest worthwhile change in 20-m sprint in the present study was 0.8%. The thresholds used in the spreadsheet for assigning the qualitative terms to chances of beneficial, trivial and harmful effects were <1% almost certainly not, 1–5% very unlikely, 5–25% unlikely 25–75% possibly, 75–95%, likely, 95–99% very likely; and >99% almost certain.[13] A beneficial or meaningful performance change was accepted when there was a more than 75% likelihood that the true value of the standardized mean difference was greater than the smallest worthwhile (beneficial) change.[13] All statistical analyses were conducted using Stata 12 (Stata Corp, College Station, TX, USA) for MacIntosh with the level of significance set at $p \leq 0.05$.

3. Results

After the sled push at 75% body mass, sprint time was 0.26 ± 1.03% slower at the 15 s time point ($p = 0.91$). There was a possibly beneficial effect of performing the sled push at the 4-min time point with a 0.95 ± 2.00% faster sprint time and a very likely beneficial effect at the 8- and 12-min time points (−1.80 ± 1.43%; $p = 0.001$ and −1.55 ± 1.54%; $p = 0.003$, respectively) (Table 1 and Fig. 2). Conversely, sprint time was slower at all the time points after the sled push at 125% body mass (1.36 ± 2.36%; $p > 0.65$–2.59 ± 2.90%; $p = 0.008$), with the likelihood of the effect being very likely harmful at the 15-s and 4-min time points and likely harmful at the 8- and 12-min time points (Table 1 and Fig. 2). Significant differences in the changes in sprint time between the sled push at 75 and 125% body mass were observed at the 15- ($p = 0.001$), 4- ($p = 0.000$), 8- ($p < 0.001$) and 12- ($p < 0.001$) min time points. The magnitude of the differences between the two sled conditions was large at the 15 s time point ($d = -1.07$) and very large at the 4- ($d = -1.431$), 8- ($d = -1.51$) and 12- ($d = -1.46$) min time points.

4. Discussion

The purpose of this study was to investigate the potentiating effect of a single sprint-style sled push stimulus loaded with either 75 or 125% body mass on subsequent unresisted 20 m sprint performance. Following the lighter sled push the data revealed a limited PAP effect at the 15 s time point but a possibly beneficial effect at

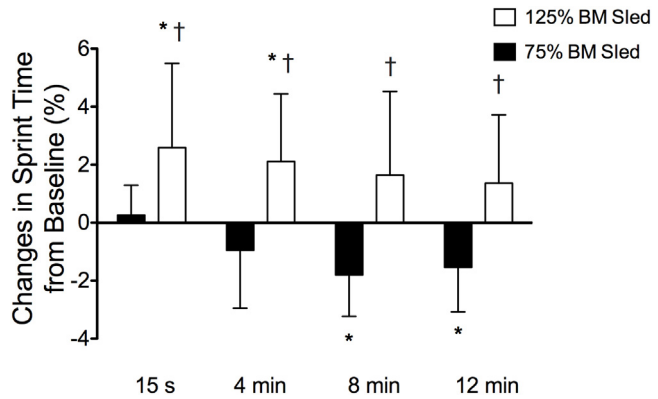

Fig. 2. Percent changes in sprint time (mean ± SD) after both the 75 and 125% body mass (BM) sled pushes. Positive values represent slower sprint times while negative values represent faster sprint time (i.e. PAP effect).
*Significantly different from baseline ($p \leq 0.05$); †significantly different from 75% Sled ($p \leq 0.05$).

the 4-min time point and a very likely beneficial effect at the 8- and 12-min time points. Conversely, slower sprint times were reported at all time points following the heavy sled push (−1.36 to −2.57%; ES = 0.38–0.67).

Although the present study is the very first to use the sled push exercise to induce a sprint PAP effect making direct comparison with the literature rather difficult, two studies have used the sled pull exercise to potentiate subsequent unresisted sprint performance.[9,10] In the study by Winwood et al.[10] rugby players performed 15 m sprints before and 4, 8 and 12 min after a sled pull loaded with either 75 or 150% body mass. Similar to the present study, the data for the light sled push condition showed faster sprint time at the 4- (ES = −0.09), 8- (ES = −0.24) and 12- (ES = −0.22) min time point indicative of a PAP effect. Conversely, sprint time was slower at all time points after the heavier sled pull stimulus (ES = 0.01–0.13). The interaction between fatigue and potentiation co-existing following a conditioning activity[8] may explain these results: sprint performance may have improved after the lighter sled push because potentiation overcame fatigue but decreased after the heavier stimulus because fatigue dominated. To note, in the present study, sprint time was mostly impaired immediately after the heavy sled push conditioning activity (+2.57 ± 2.92%) and nearly returned to baseline value by 12 min (+1.36 ± 2.36%) suggesting that fatigue culminated immediately after the sled push and then dissipated throughout the recovery period. One explanation for this is that, similar to heavy sled pulls, the heavier sled push may have induced detrimental changes in sprinting technique like increased trunk lean angle or hip joint angle.[15] These acute alterations may have resulted in acute deterioration of unresisted sprint performance leading to slower sprint times. It is worth noting that

compared to the present study Winwood et al.[10] reported slightly smaller effect sizes after the lighter sled pull (ES = 0.22–0.24) while the meta-analysis by Seitz and Haff[1] found a greater sprint PAP effect (ES = 0.51). These differences may be due to variables influencing both the occurrence and magnitude of PAP including the characteristics of both the CA (i.e. type, mechanical work etc.) and the individual (i.e. strength level, training age etc.) as well as the rest period between the prelaod exercise and the performance task.[1]

Contrary to the above-mentioned findings, Whelan et al.[9] failed to potentiate short-distance sprint velocity after three 10-m sled pulls interspersed with 90 s of recovery and loaded with 25–30% body mass in physically active males. Given that Winwood et al.[10] could potentiate sprint performance with a 75% body mass sled pull, it appears that a 25–30% body mass load may not provide enough load in order to trigger the mechanisms responsible for PAP. The ability of both the sled pull and the sled push exercises to potentiate subsequent sprint performance may be explained by the mechanical similarities between resisted sled sprinting and unresisted sprinting. Pulling or pushing a sled is a translational movement putting the body in a greater forward lean, which closely corresponds to both the start and the acceleration phases of sprinting and is likely to emphasize extension of the foot, knee, and hip.[16] Furthermore, a greater forward lean may improve the ability to apply more horizontal forces,[17] which has been associated with greater sprint acceleration than applying force in a more vertical direction.[18,19] Therefore, it could be argued that the ability of the players to apply force in the horizontal direction may have been potentiated after the lighter sled stimulus resulting in an improved acceleration and thus a faster sprint time. However, further research determining whether horizontal force production during sprinting is potentiated following a sled conditioning activity is needed to elucidate this hypothesis.

Another important finding of the present study is the existence of a responder versus non-responder phenomenon. Following the 75% body mass stimulus the percentage of players exhibiting a PAP effect at the 15-s, 4-, 8- and 12-min time point was 35%, 55%, 70% and 70%, respectively. After the sled push at 125% body mass, a PAP effect was elicited in 10, 20, 30 and 25% of the players at the 15-s 4-, 8- and 12-min time points, respectively. Furthermore, the 16 and 8 players who exhibited a PAP response after the sled push at 75% and 125% body mass, respectively, obtained their highest PAP response at either the 4-, 8- or 12-min time points. These findings might be explained by the individual's strength level that has been shown to influence both the ability to express PAP and the time necessary to realize a PAP response.[20,21] Future research should determine the PAP response of strong and weak individuals after light and heavy sled pushes stimuli.

5. Conclusion

The present findings have important training implications for practitioners seeking to utilize the sled push exercise to potentiate subsequent sprint performance. A coach seeking to potentiate 20 m sprint performance should consider using a sled loaded with 75% body mass rather than a sled loaded with 150% body mass. Moreover, a 4–12 min rest period should be observed between the sled and the sprint activity in order to benefit from PAP. Furthermore, using the present protocol may be required before implementing such technique during training since it appears that some players do not seem able to benefit from both the light and heavy sled push stimuli.

The absence of a control condition where each athlete would be required to sprint at each time point following the baseline 20 m sprint but without the sled push to see the effect on their one-off sprint performance could be considered a limitation of the study. However, the addition of a fourth session was not feasible due to the time constraints imposed by the technical staff of the team. Finally, the small effect sizes (improvement) reported in the present study could be considered a limitation, however it is worth noting that the faster sprint time at the 4-, 8- and 12-min time points after the light sled push ($-0.95 \pm 2.00\%$ to $-1.80 \pm 1.43\%$) is likely to be worthwhile for athletes requiring high levels of speed as, according to the recommendation of Hopkins,[14] coaches and sport scientists should focus on enhancements as little as 0.3–1.5% for elite athletes during event performance. Moreover, from a training perspective, even a small PAP effect elicited across multiple sets of sprints may possibly allow a greater training stimulus to occur over time,[22] resulting in greater changes in performance.

5.0.1. Practical implications

- Strength and conditioning professionals who use the sled push exercise to potentiate short sprint performance should consider the load on the sled since a 75% body mass sled push appears to induce a PAP effect while sprint performance is impaired after a sled loaded with 125% body mass.
- The sprint PAP effect is present 4–12 min following the 75% body mass sled push.
- There is a high inter-individual susceptibility in the PAP response.

Acknowledgement

There was no financial support for this study.

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